How Rationals Boost Textual Entailment Modeling: Insights from Large Language Models

Abstract

This study introduces an innovative methodology for rationale-based distillation in textual entailment. Central to our methodology is the use of diverse and deep rationale types generated by large language models, eliminating the need for explicit feature engineering between text-hypothesis pairs. Through extensive experimentation, we demonstrate the effectiveness of our rationale-enhanced distillation process, underpinned by a comprehensive study of rationales. Remarkably, our model, which utilizes the most impactful rationales and operates with 795 times fewer parameters, exhibits competitive performance, especially in contexts limited by resource availability. Specifically, our model significantly surpasses RoBERTa, FLAN-T5 Large, and GPT-3.5 in performance. Our findings underscore the potential of rationale-based approaches, which improve textual entailment modeling and pave the way for future research. These techniques could be applied to other areas of NLP and beyond. This study's contributions are poised to benefit researchers and practitioners seeking to leverage the power of large language models in augmenting reasoning capabilities with rationales. The novelty of our approach lies in its ability to deliver high performance with significantly fewer computational resources, making it a valuable advancement in the field.

Keywords: Rationale, Distillation, Large Language Model, Textual Entailment

1 Introduction

In the rapidly advancing field of natural language processing (NLP), the capability of computational systems to interpret human language forms a fundamental foundation, underpinning a multitude of applications. The goal of language understanding systems is to explore and digest the meaning of tokens and their interrelationships. Among various NLP tasks, textual entailment (TE) stands out as a crucial task, particularly in understanding the relationships between words. This task is concisely defined in seminal works [1, 2], where the primary objective is to deeply comprehend and elucidate the relationship between two textual entities: premise `T` and hypothesis `H`. The fundamental challenge is for the model to accurately determine whether the hypothesis can be logically inferred from the premise. TE provides essential insights into the mechanisms underlying language comprehension and reasoning, involving the determination of directional relations between text fragments. The process involved in TE requires not only the representation and understanding of a single sentence but also the determination of linguistic links between them.

Building upon the above discussion of the challenges and fundamental role of TE in language understanding, its wide-ranging applications significantly impact various aspects of NLP. TE's most notable application lies in enhancing the accuracy of question-answering (QA) systems [3, 4]. The integration of TE helps refine the process of understanding and responding to queries in QA systems, yielding more accurate and contextually relevant answers. Furthermore, TE plays a pivotal role in other domains such as machine translation [5], and knowledge graph completion [6]. The ubiquitous presence of TE across these key areas of NLP underscores its vital importance. TE not only lays the groundwork for complex tasks of language understanding but also advances the field of NLP towards more sophisticated and human-like language processing capabilities.

A significant challenge in the deep learning approaches to this task lies in its dependency on data. To build an effective model, previous methods heavily relied on gold-standard datasets, which are laboriously annotated in the manual. While these datasets offer high-quality training samples, their high cost and limited quantity present obstacles to developing TE models across various domains. Recently, the advantage of transfer learning and self-supervised learning has marked a paradigm shift in language processing models. Semantic features of texts are now extracted based on pre-trained language models utilizing large-scale, self-generated data. This advancement has led to a more efficient representation of textual context. Nonetheless, during the classification phase aimed at identifying the relationship between a context sentence and its hypothesis, deep learning models still depend on supervised datasets that are costly and limited in quantity.

Despite these advances, there remain specific gaps that this research aims to address. Current methodologies often struggle with scalability and generalizability across diverse linguistic datasets due to their reliance on expensive, annotated data. Additionally, there is a lack of integration of rationale-based methods that could potentially offer more explainable and efficient models. These gaps highlight the need for novel approaches that can leverage the capabilities of large language models while minimizing resource dependency.

Recently, the emergence and rapid proliferation of large language models (LLMs) [7] trained on vast amounts of data, have marked exceptional skills across diverse tasks, showcasing their reasoning in both zero-shot and few-shot learning. Instead of grappling with the costly expansion of gold-standard datasets, our system focuses on harnessing the condensed knowledge within these large language models to enrich our resources for this task. Indeed, self-generated knowledge from large language models is abundantly available and easily accessible. To utilize the self-generated data, we propose a novel training technique that leverages the power of large language models by integrating the rationales of text-hypothesis pairs as additional supervision. This approach holds great potential in enhancing effectiveness with the available data and capitalizing on the surge of large language models' capabilities.

Our methodology uniquely integrates the strengths of multi-task learning by combining the decision-making process with explanations, presenting a more comprehensive approach to TE. Initially, our model leverages a large language model to generate justifications, clarifying the entailment relationships between premises and hypotheses. Following this, we propose a system designed to integrate this generated information, fully leveraging the advantages of multi-task learning. Through a comprehensive analysis of various types of rationales, our study highlights their crucial role and transformative impact on textual entailment, marking a significant potential from traditional methodologies. The thorough examination allowed us to identify and utilize rationales that most significantly enhance model comprehension and effectiveness.

Our approach addresses challenges in processing languages with scarce resources and pioneers innovative methods to boost the efficiency and effectiveness of machine learning models in multi-modal contexts. The primary contributions of our research are as follows:

- We introduce a novel methodology for rationale-based distillation in TE, emphasizing the diversity and depth of rationale types studied. This technique improves the model's effectiveness without necessitating explicit feature engineering between text-hypothesis pairs.
- The effectiveness of our rationale-enhanced distillation process, underpinned by our comprehensive rationale study, has been proven through extensive experimentation. Using the most impact rationales, the model with 795 times fewer parameters, shows competitive performance, particularly in resource-limited language contexts.

The structure of this paper is organized as follows: Section 2 delves into related research, focusing on the capabilities of LLMs and the process of learning through reasoning distillation from LLMs. Section 3 outlines our methodology, detailing the approach we have adopted. Section 4 presents the experiments conducted, alongside an analysis of our findings. Finally, Section 5 concludes our study, summarizing the key outcomes and suggesting directions for future research.

2 Related Works

2.1 Large language models for NLP tasks

Large language models have emerged as a tour de force in various NLP tasks, showcasing impressive capabilities [8–12]. A noteworthy emerging property of LLMs is their ability to generate supportive rationales for predictions. Recent studies have focused on eliciting this reasoning capability from LLMs [11, 13, 14] and utilizing this knowledge to train more efficient downstream models. The concept of chain-of-thought (CoT) reasoning, introduced by Wei et al. [11], has been pivotal in enhancing the reasoning capabilities of LLMs. It involves augmenting input prompts with reasoning steps to improve performance in few-shot or zero-shot scenarios and subsequently employing the "self-improve" method to utilize these rationales as additional data sources [15, 16]. To improve LLM adaptation to downstream tasks and enhance generalization to unseen classes, Wang et al. [17] employ instance-level information and task-level information to tune prompts. Despite their prowess, deploying LLMs in environments where offline access and privacy are paramount remains a challenge. To address these limitations, we developed a smaller model derived from the knowledge generated by LLM.

2.2 Learning with reasoning distillation from LLMs

While utilizing LLM-generated rationales represents an exciting research avenue, the use of human-generated rationales has also proven effective [18, 19]. Human rationales can serve various functions: guiding model predictions [20], regularizing model behavior [21], enhancing overall model performance [22–26], and even acting as standard labels for models to emulate. However, generating human labels is both time-consuming and costly. Recent efforts have explored distilling the reasoning ability of LLMs into smaller language models (LMs), where this ability is seen as an emergent property that enhances performance in reasoning tasks through CoT. This approach involves leveraging rationales generated by LLMs to train smaller, task-specific models, thereby improving their performance without incurring substantial computational or memory costs [27]. Research employing extracted rationales for supervision [28–32] has demonstrated potential, albeit with certain constraints. For instance, PINTO [33] uses LLM to generate rationales during test time, which does not entirely resolve deployment challenges. In contrast, the distilling step-by-step approach [32] offers a more comprehensive solution by utilizing LLM-generated rationales as additional, richer information to train small models through a multitask training setting. Furthermore, recent advancements have showcased significant performance enhancements through innovative rationale utilization strategies. Distinctly, some studies have shifted from generating multiple rationales with a single model to employing various large language models (LLMs) to produce a diverse set of rationales, each from a different model perspective [34]. While innovative, this multi-model rationale generation approach introduces substantial challenges in resource allocation and deployment. Moreover, it raises concerns regarding the consistency in the quality of rationales, as ensuring uniform quality across different LLMs remains a significant hurdle. Similarly, another method uses an LLM to assess and refine the outputs from smaller models critically [35]. Both approaches, although pioneering, share common limitations regarding the substantial resources needed for deployment and the complexity involved in integrating LLMs with smaller models. These limitations underscore the need for more efficient integration techniques that mitigate resource intensity and address quality consistency, presenting a critical area for future research and development.

Building on this analysis, our work aims to provide deeper insights into the efficient use of rationales with varying training datasets. We explore combining multiple types of rationales to enhance the performance of smaller models.

3 Methodology

Our proposed framework, as depicted in Figure 1, capitalizes on these capabilities. It encompasses two primary steps: initially, an LLM is employed with an unlabeled dataset to generate output labels accompanied by rationales - these are natural language explanations justifying the predicted labels. Subsequently, these rationales, along with the task labels, are utilized to train smaller downstream models. Rationales provide a deeper understanding of the reasoning behind an input's classification, often encapsulating crucial task-specific knowledge that may not be immediately discernible from the inputs alone.

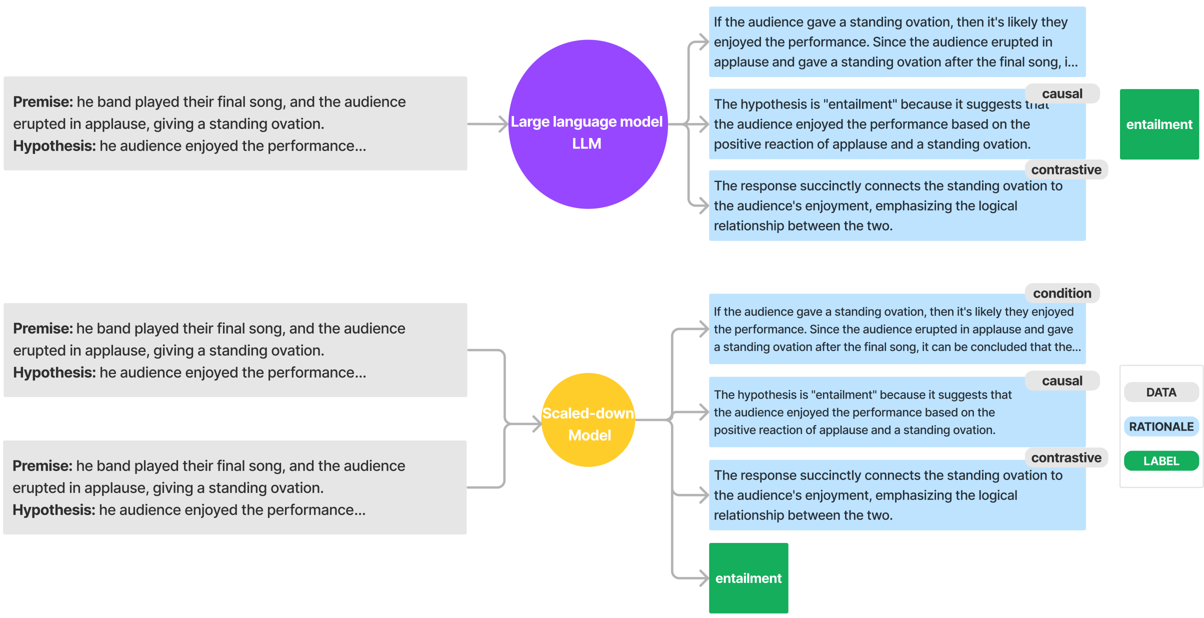

Fig. 1: Our model proposed to boost a distilled model via llm-generated rationales

3.1 Refinement of the distilling step-by-step approach

Building upon the innovative "Distilling Step by Step" methodology from existing literature, this study seeks to harness the reasoning prowess of large language models (LLMs) for training smaller models in a data-efficient manner. The process unfolds in two fundamental steps. Initially, an LLM is utilized to generate output labels complemented by justifying rationales. These rationales, articulated in natural language, serve as supportive explanations for the model's predicted label, as depicted in the accompanying figure. The subsequent step involves repurposing these rationales as task-specific labels to train smaller, downstream models.

This method has demonstrated superiority over traditional fine-tuning and standard distillation across various tasks. Informed by this significant breakthrough, our research delves further into multiple facets of the method. Firstly, we scrutinize the impact of different types of rationales on the training efficiency of smaller models, seeking to identify the most effective rationale. Secondly, we experiment with the amalgamation of various rationales, a strategy that has yielded remarkable performance enhancements. This exploration aims to optimize the training process and maximize the efficacy of smaller models using a diverse and integrated rationale approach.

3.2 Distilling step by step with multiple types of rationales.

3.2.1 Extracting rationales from LLMs

Recent studies have shed light on a particularly notable capability of large language models (LLMs) – their ability to generate rationales that bolster their predictions [11, 14]. Focusing on this emergent property, several researchers have endeavored to harness LLMs' reasoning capacity, utilizing these generated rationales to train smaller, more efficient models.

In our approach, we employ the chain-of-thought (CoT) prompting method [11] to provoke and extract rationales from LLMs, as exemplified in figure 1. This process involves presenting the LLM with a set of premises and hypotheses, prompting it to generate not only answers but also the underlying rationales for those answers. Each input is formatted as a pair (question, choices) to stimulate the LLM into producing both rationales and corresponding labels for each query.

Compared to human labels, the generated rationales have better structure and meaning, leading to better performance. Previous work used rationales to increase performance but did not go into detail about each type of influence [32]. In contrast, this article will provide a more detailed examination of the different kinds of rationales and their impacts. Specifically, the performance of models using LLM-generated rationales is approximately 11.23% higher than those using traditional methods.

While the responses from LLMs may seem adequate, our study extends to examining how different types of rationales impact the training and performance of smaller downstream models. To this end, we present each question to the LLMs along with seven distinct types of rationales: conditional (explain the reasoning using conditional statements), contrastive (highlight what sets the choice apart from the others), neutral (provide a straightforward explanation), consensus (justify the answer based on general knowledge), causal (explain the cause-and-effect relationship between the question and the chosen answer), comparative (explain why the choice is better than the others), and historical (provide historical context in the rationale). This comprehensive approach aims to discern which rationale types most effectively contribute to the enhanced performance of smaller models in downstream applications.

3.2.2 Training compact models using rationales in the text-to-text framework

This section discusses our approach to enhancing the training of task-specific models within the text-to-text framework by incorporating rationales into the learning process. Originally developed for a broad range of NLP tasks, this framework has been modified to integrate rationale-based learning, offering a more detailed method of understanding model decisions.

Background and previous methods: conventionally, task-specific models are fine-tuned using supervised data on pre-trained models [36]. Human-annotated labels are resource-intensive, leading to the adoption of task-specific distillation techniques [37, 38]. These methods utilize large language models (LLMs) to create pseudo-labels, $\hat{y}_i$, as a substitute for human annotations [39–41].

The objective for smaller models in all these cases is to minimize the discrepancy between predicted and actual labels, represented by the loss function:

$$\mathcal{L}_{\text{label}} = \frac{1}{N}\sum_{i=1}^{N} \ell(f(x_i), y_i), \tag{1}$$

where ℓ denotes the cross-entropy loss between predicted and target tokens. We assume human-annotated labels, y_i, for standard fine-tuning and LLM-predicted labels, $\hat{y}_i$, for distillation.

Incorporating rationales through multi-task learning: building on Eq. (1), we aim to establish a clearer link between inputs x_i and outputs y_i by incorporating rationales r_i as an additional layer of supervision. Previous studies introduced rationales as an extra input [20, 33], training the model on both text and rationales (x, r):

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N} \ell(f(x_i, r_i), y_i). \tag{2}$$

However, this approach necessitates an LLM to generate rationales before prediction, limiting its deployability due to the ongoing need for an LLM.

Step-by-Step distillation with rationales: in our novel approach, rationales are integrated as part of a multi-task learning framework. Rather than using rationales as direct inputs, we train the model to produce both the task labels and corresponding rationales from the text inputs, $f(x_i) \rightarrow (y_i, r_i)$:

$$\mathcal{L} = \lambda\mathcal{L}_{\text{label}} + (1-\lambda)\mathcal{L}_{\text{rationale}}, \tag{3}$$

where $\mathcal{L}_{\text{label}}$ is the label prediction loss from Eq. (1), and $\mathcal{L}_{\text{rationale}}$ represents the rationale generation loss:

$$\mathcal{L}_{\text{rationale}} = \frac{1}{N} \sum_{i=1}^{N} \ell(f(x_i, r_i), y_i). \tag{4}$$

The rationale generation loss Eq. (4) helps the model learn to generate intermediate reasoning steps, thereby enhancing its ability to predict the final label. Importantly, unlike the method outlined in Eq. (2), this approach does not require rationales at test time, removing the dependency on an LLM during deployment.

Diversifying rationales with step-by-Step distillation: recognizing that a single question may yield multiple valid responses, our methodology aims to enrich the training process by incorporating a variety of rationales. This diversity in rationales bolsters the link between input data x_i and output results y_i. Drawing inspiration from recent studies on open-ended generation, such as the "Self-consistency for open-ended generations" paper, we utilize an array of rationales to optimize the likelihood of deriving the most accurate answer from different rationale types.

To implement this, we incorporate "task prefixes" ([label], [rationale]) into our input data. This method conditions the model to produce the desired output y_i when presented with the [label] prefix, and to generate the corresponding rationale r_i when provided with [rationales]. This technique, as outlined by Raffel et al. [42], not only deepens the model's comprehension of the data but also introduces greater flexibility and depth to the training process. By leveraging diverse rationales, we enable the model to approach a problem from multiple perspectives, thus enhancing its ability to handle a wide range of tasks and improving its overall performance.

3.3 Enhancing model efficacy through step-by-step distillation with multiple rationales

Building upon the previously discussed distilling step-by-step method, our objective is to forge stronger connections between input data and output labels. This approach is distinguished by its utilization of a diversified set of rationales, coupled with the absence of large language models (LLMs) during the testing phase. We refine our model training process, $f(x) \to (y_i, r_i)$, to not only generate a single rationale but also multiple rationales concurrently based on the text inputs:

$$\mathcal{L}_{\text{rationale}} = \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}_{\text{rationale}_i}. \tag{5}$$

Here, Eq. (5) $\mathcal{L}_{\text{rationale}}$ represents the collective rationale generation loss, as introduced in Eq. (3), and $\mathcal{L}_{\text{rationale}_i}$ signifies the rationale generation loss for each specific type of rationale.

This enhancement in the rationale generation loss mechanism enables the model to not only generate a singular rationale but also to simultaneously produce multiple rationale types. An important aspect of this approach is its capacity to discern and learn the interrelations among various rationales. This comprehensive understanding significantly augments the model's predictive accuracy and overall performance. By incorporating multiple rationales, the model gains a more holistic understanding of the data, allowing it to make more informed and nuanced predictions.

4 Experiments

4.1 Dataset

To evaluate the contribution of rationales on boosting reasoning learning, the experiments are conducted using two renowned benchmark datasets. For textual entailment, we utilize e-SNLI [24], and for commonsense question answering, we employ CQA [20, 43]. It's essential to note that, to the best of our understanding, these datasets are free from personally identifiable information or offensive content.

- **e-SNLI**: Developed by Camburu et al. [24], e-SNLI augments the original SNLI dataset with over 570,000 (549,367 train, 9,842 validation, 9,824 test) human-annotated examples, adding explanations to support entailment, contradiction, and neutral decisions in natural language inference tasks. It is a valuable asset for NLP applications, available via the Hugging Face datasets repository. https://huggingface.co/datasets/esnli
- **CQA**: Introduced by Talmor et al. [43], this question-answering dataset challenges AI with commonsense reasoning. It includes 8,766 training, 975 validation, and 1,221 test examples, each enriched with human explanations to foster deeper understanding. Available for public use, CommonsenseQA is key for advancing AI's grasp of everyday knowledge. https://www.tau-nlp.sites.tau.ac.il/commonsenseqa

For the e-SNLI dataset, we utilize the original training and validation sets. Due to the unavailability of ground-truth labels for the CQA dataset for the original test set, we employ the original validation set as our test set. The dataset statistics are presented in Table 1.

Dataset	Train	Validation	Test
e-SNLI	549,367	9,842	9,824
CQA	8,766	975	1,221

Table 1: Dataset statistics used in our experiments.

4.2 Experimental settings and results

In our experimental setup, we deviate from the methodology outlined in the original study by utilizing the 175B-parameter GPT-3.5 model, in contrast to the larger 540B-parameter PaLM model previously employed. For tasks that require specific modeling approaches, we incorporate T5 models equipped with pre-trained weights sourced from Hugging Face's repository. Our approach to prompting leverages the OPENAI

API to systematically generate chain of thought (CoT) rationales through structured prompts. Our experiments were conducted on GPU P100 instances provided by the Kaggle platform, leveraging the following pre-trained models:

- **T5-Base (200M parameters)**: Configurations for this model were adapted from the Hugging Face Transformers library, with a learning rate of $= 5 \times 10^{-5}$, a batch size of $= 8$, max input length $= 1024$, and a training duration of up to 4000 steps.[1]
- **RoBERTa Base (125M parameters)**: We utilized the version available on the Hugging Face repository.[2]
- **FLAN-T5 Large (783M parameters)**: This model was obtained from Google's public repository on Hugging Face.[3]

As shown in Table 2, the proposed method, enhanced by rationales, outperforms ChatGPT 3.5 across both ESNLI and CQA tasks despite utilizing 795 times fewer parameters. This impressive performance demonstrates the efficiency and effectiveness of our approach. Specifically, RoBERTa Base and FLAN-T5 Large, even after fine-tuning, achieve only 44.08% and 57.29% accuracy on the ESNLI task, respectively. On the other hand, the distilling step-by-step method reaches 75.38% accuracy, which is significantly higher but still falls short of our method's performance. Our method achieves an accuracy of 83.93% on the ESNLI task, substantially outperforming all other models listed. In the CQA task, FLAN-T5 Large achieves 60.11% accuracy, while the distilling step-by-step method reaches 55.45%. ChatGPT 3.5 leads with 77.23% accuracy, yet our method remains competitive with an accuracy of 60.36%, using far fewer parameters.

The effectiveness of our method is primarily attributed to its strategic emphasis on different kinds of rationales. Unlike approaches that allow language models (LLMs) to respond in their inherent style, our method strictly guides responses to adhere to specific rationale types, such as conditional, contrastive, and neutral, among others. This targeted approach not only improves the model's learning efficiency but also deepens its understanding of the underlying rationale nuances, leading to enhanced performance and insights. In the following section, we delve into a detailed discussion on which type of rationale yields robust performance.

In summary, our method demonstrates that with a focused and rationale-guided approach, it is possible to achieve superior performance even with a significantly smaller model size. This advancement highlights the potential for more efficient and scalable AI models in practical applications.

4.3 Rationale type comparison

This section examines how different rationale types influence performance, compared to the standard Distilling step-by-step method. In Figure 2, our analysis of the e-SNLI dataset shows a uniform enhancement in performance across all rationale types when compared to the baseline, with neutral rationales achieving a notable 83.93% accuracy and causal rationales close behind at approximately 81.92% while the baseline

[1] Hugging Face Transformers
[2] Project Aina RoBERTa Base.
[3] Google FLAN-T5 Large

Model	ESNLI ↑ (%)	CQA ↑ (%)
RoBERTa Base (125M)[44]	44.08	-
FLAN-T5 Large (783M) [45]	57.29	60.11
Standard Finetune T5 Base (220M) [46]	61.41	52.74
ChatGPT 3.5 (175B)	72.99	77.23
Distilling Step by Step (220M) [47]	75.38	55.45
Our method (220M)	**83.93**	**60.36**

Table 2: Comparison of model performance. The results are statistically significant ($p < 0.05$) via the pairwise t-test.

just approximately 75.38%. This suggests that these rationale types are particularly effective for textual entailment.

Conversely, the performance on the CQA dataset is more variable. While neutral rationales still lead with around 60% accuracy, condition, and comparative rationales fall short of the baseline, scoring near 52% and 54% respectively. This performance dip indicates that these rationale types may not be as suitable for the intricacies of commonsense question answering, where a direct comparison or conditional reasoning might not always apply. These findings emphasize the need for task-specific rationale selection to optimize model performance.

4.4 Error Analysis

To dive into the error analysis, let's examine the performance of our model compared to the LLM. Our model achieved 83.93% accuracy for neutral-type rationales, while the LLM reached only 72.99%. This demonstrates that our model can effectively learn from the rationales provided in the questions.

For instance, consider the first example in Table 3 where the premise is "A man wearing a gray sweater walking through a pile of leaves" and the hypothesis is "The man is outside in the Fall.". In this case, the correct label is "neutral". However, the LLM incorrectly labels it as "entailment". This means that the hypothesis is necessarily true given the premise. A man wearing a gray sweater walking through a pile of leaves implies that he is outside in the Fall." Here, the LLM infers the answer incorrectly. Despite the incorrect rationale input, our model accurately identifies the label as 'neutral'. This demonstrates our model's ability to understand and correctly answer questions, enhancing its performance over the LLM. Specifically, our model's performance is 1.15 times that of the LLM, indicating significant development potential in the near future.

However, our model also has instances of underperformance. For instance, consider the third example in Table 3, where the premise is "The boy in the blue and yellow top is standing with arms outstretched" and the hypothesis is "The boy in the blue and yellow top is standing with arms outstretched 40 feet wide.". The correct label is "contradiction". The LLM correctly identifies this, stating, "The correct answer is contradiction. This means that the hypothesis is false given the premise. The premise states that the boy is standing with arms outstretched, which implies a normal human arm span. The hypothesis that the boy is standing with arms outstretched 40 feet wide is absurd and contradicts the premise.". In contrast, our model incorrectly labels this

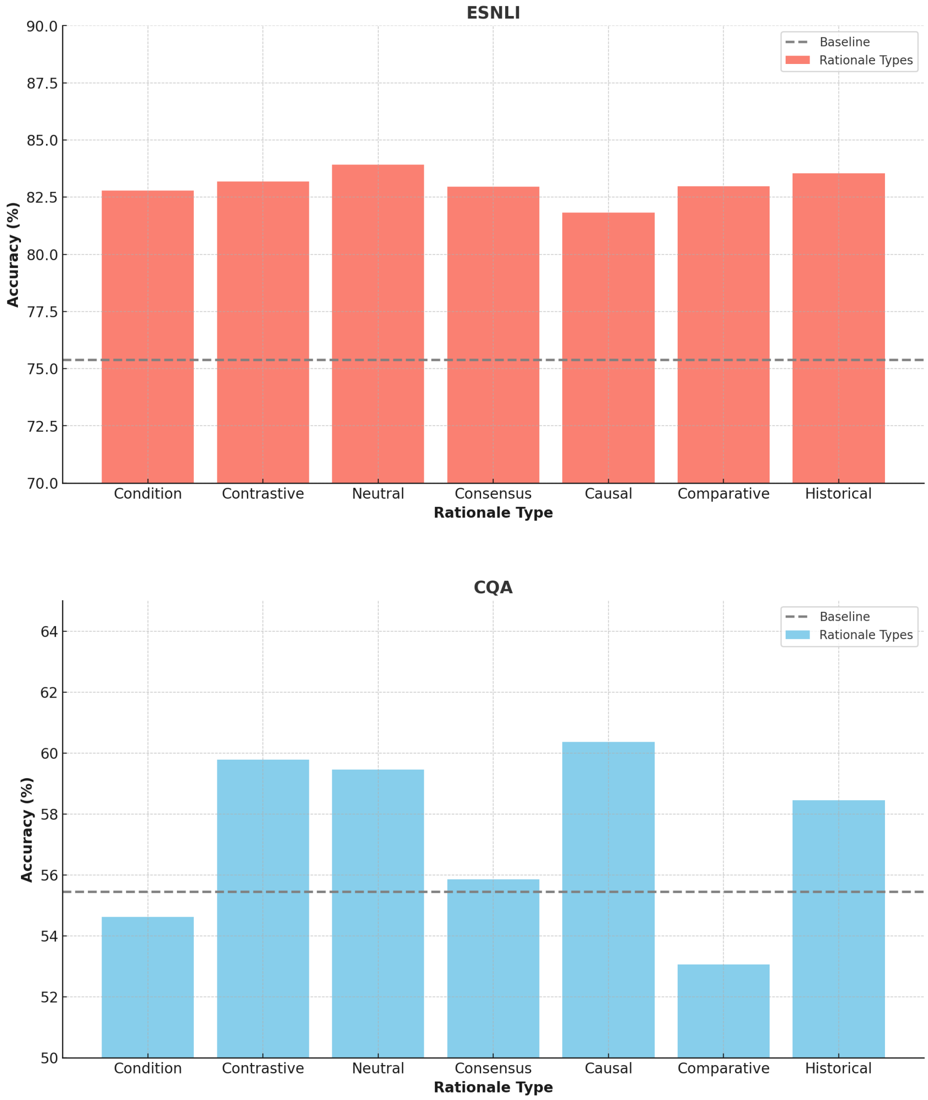

Fig. 2: Performance comparison of various rationale types against the standard distilling step-by-step Method. The graph illustrates the differential impact of each rationale type on model performance, with notable improvements on the e-SNLI dataset and varied results on the CQA dataset, underscoring the importance of rationale selection.

as "neutral". Although our model does not have prior knowledge about human arm span like the LLM, it avoids labeling the hypothesis as "entailment". This indicates some level of understanding and decision-making ability, though the answer is still incorrect.

Premise	**Hypothesis**	**Label**	**LLM Answer**	**Scaled-down Label**
A man wearing a gray sweater walking through a pile of leaves	The man is outside in the Fall.	neutral	The correct answer is **entailment**. This means that the hypothesis is necessarily true given the premise. A man wearing a gray sweater walking through a pile of leaves implies that he is outside in the Fall.	neutral
Two children, in colorful outfits, playing in a field with a big rock in the middle.	Two kids are playing with a big rock in the field with family.	neutral	The correct answer is **contradiction**. The hypothesis adds an extra detail that is not present in the premise. The premise contradicts the hypothesis.	neutral
The boy in the blue and yellow top is standing with arms outstretched.	The boy in the blue and yellow top is standing with arms outstretched 40 feet wide.	contradiction	The correct answer is **contradiction**. This means that the hypothesis is false given the premise. The premise states that the boy is standing with arms outstretched, which implies a normal human arm span. The hypothesis that the boy is standing with arms outstretched 40 feet wide is absurd and contradicts the premise.	neutral
Woman in white in foreground and a man slightly behind walking with a sign for John's Pizza and Gyro in the background.	A man is advertising for a restaurant.	entailment	The correct answer is **entailment**. This means that the hypothesis is true given the premise. The premise shows a man walking with a sign for John's Pizza and Gyro, which implies that he is advertising for a restaurant.	neutral

Table 3: Premises, Hypotheses, Labels, LLM Answers, and Scaled-down Labels

In summary, while our model demonstrates superior performance in certain cases by learning from rationales, it also exhibits weaknesses in specific instances where additional contextual knowledge is required. This highlights areas for further improvement and refinement to enhance our model's overall accuracy and robustness.

4.5 How many rationales are suitable

In this section, we analyze the effects of utilizing multiple rationales in our methodology, as detailed in Table 4. Our findings indicate that leveraging two rationales (contrastive and historical) simultaneously enhances the model's performance marginally, increasing it from 83.93% to 84.62%. This demonstrates the potential of multi-rational learning to refine performance beyond what is achievable with a single rationale. However, our results also reveal a limitation in this approach: incorporating three or four rationales simultaneously does not sustain this positive trend and leads to overfitting. This suggests that while there is value in integrating multiple rationales, there is a threshold beyond which the complexity introduced outweighs the benefits, detracting from the model's generalizability and effectiveness.

Model	Performance ↑ (%)
Distilling Step by Step (220M)	75.38
Single Rationales (220M)	83.93
Two Rationales (220M)	**84.62**
Three Rationales (220M)	**81.80**
Four Rationales (220M)	**80.92**

Table 4: Model performance of multiple rationales. The results are statistically significant ($p < 0.05$) via the pairwise t-test.

5 Discussion

We propose a method called Distilling Step-by-Step with Multiple Rationales for extracting rationales from large language models (LLMs) to provide informative supervision in training small task-specific models. Our results demonstrate that this approach outperforms both traditional LLMs and other methods that use only rationales. Moreover, it reduces the need for extensive training datasets by replacing human-generated rationales. Distilling Step-by-Step proposes a resource-efficient training-to-deployment paradigm compared to existing methods. Further studies affirm the generalizability and the strategic design choices of our method. Below, we discuss the limitations, future directions, and ethical considerations of our work.

Our approach has several limitations. Firstly, it requires the use of the GPT-3.5 API, which is not only limited but also incurs costs. Secondly, although rationales significantly enhance the performance of our task-specific models, the evaluation of a rationale's quality whether it is effective or not remains unclear. Finally, despite the success of using LLM rationales, there is evidence that LLMs exhibit limited reasoning capabilities in more complex reasoning and planning tasks [48].

Looking ahead, there are several directions for future work with our method. First, we could explore the use of other LLMs, such as GPT-4.0 or Gemini, to potentially improve the quality of rationales, anticipating that newer models might offer enhanced capabilities. Second, we aim to investigate how the quality of rationales impacts the performance of task-specific models [49]. Finally, while we have observed success in using LLM rationales for textual entailment tasks, we plan to extend this method to tasks like image description to enhance both the classification and description of images using similar strategies.

It is important to acknowledge that the behavior of our smaller downstream models may inherit biases from the larger teacher LLMs. We anticipate that advancements in reducing antisocial behaviors in LLMs could be adapted to enhance the ethical performance of smaller language models.

6 Conclusion

Our research introduces a novel approach to textual entailment by ingeniously integrating the strengths of multi-task learning with a rationale provision process. This methodology not only leverages the generative capabilities of large language models to produce justifications that clarify entailment relationships but also incorporates these justifications into a system designed to exploit multi-task learning benefits fully.

Our in-depth exploration of various rationale types underscores their vital importance and revolutionary impact on understanding and improving textual entailment, setting our work apart from conventional methods. The comprehensive analysis conducted has enabled us to identify and apply the most effective rationales, significantly enhancing model comprehension and performance.

The success of our rationale-enhanced distillation approach is demonstrated through extensive testing, showcasing that our model, despite having 795 times fewer parameters, achieves competitive performance in both of textual entailment and QA tasks. Specifically, our model achieves up to 1.3 times higher accuracy compared to previous approaches. This method offers significant potential for future investigations in diverse areas to enhance reasoning abilities with the integration of rationales.

References

[1] Dagan, I., Glickman, O., Magnini, B.: The pascal recognising textual entailment challenge. In: Machine Learning Challenges Workshop, pp. 177–190 (2005). Springer

[2] Sharma, N., Sharma, R., Biswas, K.K.: Recognizing textual entailment using dependency analysis and machine learning. In: Proceedings of the 2015 Conference of the North American Chapter of the Association for Computational Linguistics: Student Research Workshop, pp. 147–153 (2015)

[3] Pathak, A., Manna, R., Pakray, P., Das, D., Gelbukh, A., Bandyopadhyay, S.: Scientific text entailment and a textual-entailment-based framework for cooking domain question answering. Sādhanā **46**, 1–19 (2021)

[4] Khot, T., Sabharwal, A., Clark, P.: Scitail: A textual entailment dataset from science question answering. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 32 (2018)

[5] Padó, S., Galley, M., Jurafsky, D., Manning, C.D.: Robust machine translation evaluation with entailment features. In: Proceedings of the Joint Conference of the 47th Annual Meeting of the ACL and the 4th International Joint Conference on Natural Language Processing of the AFNLP, pp. 297–305 (2009)

[6] Yoshikawa, M., Mineshima, K., Noji, H., Bekki, D.: Combining axiom injection and knowledge base completion for efficient natural language inference. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 33, pp. 7410–7417 (2019)

[7] Chowdhery, A., Narang, S., Devlin, J., Bosma, M., Mishra, G., Roberts, A., Barham, P., Chung, H.W., Sutton, C., Gehrmann, S., Schuh, P., Shi, K., Tsvyashchenko, S., Maynez, J., Rao, A., Barnes, P., Tay, Y., Shazeer, N., Prabhakaran, V., Reif, E., Du, N., Hutchinson, B., Pope, R., Bradbury, J., Austin, J., Isard, M., Gur-Ari, G., Yin, P., Duke, T., Levskaya, A., Ghemawat, S., Dev, S., Michalewski, H., Garcia, X., Misra, V., Robinson, K., Fedus, L., Zhou, D., Ippolito, D., Luan, D., Lim, H., Zoph, B., Spiridonov, A., Sepassi, R., Dohan, D., Agrawal, S., Omernick, M., Dai, A.M., Pillai, T.S., Pellat, M., Lewkowycz, A., Moreira, E., Child, R., Polozov, O., Lee, K., Zhou, Z., Wang, X., Saeta, B., Diaz, M., Firat, O., Catasta, M., Wei, J., Meier-Hellstern, K., Eck, D., Dean, J., Petrov, S., Fiedel, N.: Palm: Scaling language modeling with pathways. Journal of Machine Learning Research **24**, 240–1240113 (2023)

[8] Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J.D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., *et al.*: Language models are few-shot learners. Advances in neural information processing systems **33**, 1877–1901 (2020)

[9] Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., Casas, D., Hendricks, L.A., Welbl, J., Clark, A., Hennigan, T., Noland, E., Millican, K., Driessche, G., Damoc, B., Guy, A., Osindero, S., Simonyan, K., Elsen, E., Vinyals, O., Rae, J.W., Sifre, L.: An empirical analysis of compute-optimal large language model training. In: Oh, A.H., Agarwal, A., Belgrave, D., Cho, K. (eds.) Advances in Neural Information Processing Systems (2022). https://openreview.net/forum?id=iBBcRUlOAPR

[10] Chowdhery, A., Narang, S., Devlin, J., Bosma, M., Mishra, G., Roberts, A., Barham, P., Chung, H.W., Sutton, C., Gehrmann, S., *et al.*: Palm: Scaling language modeling with pathways. Journal of Machine Learning Research **24**(240), 1–113 (2023)

[11] Wei, J., Wang, X., Schuurmans, D., Bosma, M., ichter, b., Xia, F., Chi, E., Le, Q.V., Zhou, D.: Chain-of-thought prompting elicits reasoning in large language models. In: Koyejo, S., Mohamed, S., Agarwal, A., Belgrave, D., Cho, K., Oh, A. (eds.) Advances in Neural Information Processing Systems, vol. 35, pp. 24824–24837. Curran Associates, Inc., ??? (2022). https://proceedings.neurips.cc/paper_files/paper/2022/file/9d5609613524ecf4f15af0f7b31abca4-Paper-Conference.pdf

[12] Sakirin, T., Said, R.B.: User preferences for chatgpt-powered conversational interfaces versus traditional methods. Mesopotamian Journal of Computer Science **2023**, 24–31 (2023)

[13] Nye, M., Andreassen, A.J., Gur-Ari, G., Michalewski, H., Austin, J., Bieber, D., Dohan, D., Lewkowycz, A., Bosma, M., Luan, D., et al.: Show your work: Scratchpads for intermediate computation with language models. arXiv preprint arXiv:2112.00114 (2021)

[14] Kojima, T., Gu, S.S., Reid, M., Matsuo, Y., Iwasawa, Y.: Large language models are zero-shot reasoners. Advances in neural information processing systems **35**, 22199–22213 (2022)

[15] Zelikman, E., Wu, Y., Mu, J., Goodman, N.: Star: Bootstrapping reasoning with reasoning. Advances in Neural Information Processing Systems **35**, 15476–15488 (2022)

[16] Huang, J., Gu, S.S., Hou, L., Wu, Y., Wang, X., Yu, H., Han, J.: Large language models can self-improve. arXiv preprint arXiv:2210.11610 (2022)

[17] Wang, H., Shao, M., Tan, X., Zhang, L.: Modal-aware prompt tuning with deep adaptive feature enhancement. Computers and Electrical Engineering **117**, 109270 (2024) https://doi.org/10.1016/j.compeleceng.2024.109270

[18] Wang, G., Jiang, X., Liu, N., Xu, X.: Language-enhanced object reasoning networks for video moment retrieval with text query. Computers and Electrical Engineering **102**, 108137 (2022) https://doi.org/10.1016/j.compeleceng.2022.

108137

[19] Shi, W., Wang, H., Lou, X.: Multi-modal graph reasoning for structured video text extraction. Computers and Electrical Engineering **107**, 108641 (2023) https://doi.org/10.1016/j.compeleceng.2023.108641

[20] Rajani, N.F., McCann, B., Xiong, C., Socher, R.: Explain yourself! leveraging language models for commonsense reasoning. arXiv preprint arXiv:1906.02361 (2019)

[21] Ross, A.S., Hughes, M.C., Doshi-Velez, F.: Right for the right reasons: Training differentiable models by constraining their explanations. arXiv preprint arXiv:1703.03717 (2017)

[22] Zaidan, O., Eisner, J., Piatko, C.: Using "annotator rationales" to improve machine learning for text categorization. In: Human Language Technologies 2007: The Conference of the North American Chapter of the Association for Computational Linguistics; Proceedings of the Main Conference, pp. 260–267 (2007)

[23] Zhang, Y., Marshall, I., Wallace, B.C.: Rationale-augmented convolutional neural networks for text classification. In: Proceedings of the Conference on Empirical Methods in Natural Language Processing. Conference on Empirical Methods in Natural Language Processing, vol. 2016, p. 795 (2016). NIH Public Access

[24] Camburu, O.-M., Rocktäschel, T., Lukasiewicz, T., Blunsom, P.: e-snli: Natural language inference with natural language explanations. Advances in Neural Information Processing Systems **31** (2018)

[25] Hancock, B., Bordes, A., Mazare, P.-E., Weston, J.: Learning from dialogue after deployment: Feed yourself, chatbot! arXiv preprint arXiv:1901.05415 (2019)

[26] Pruthi, D., Bansal, R., Dhingra, B., Soares, L.B., Collins, M., Lipton, Z.C., Neubig, G., Cohen, W.W.: Evaluating explanations: How much do explanations from the teacher aid students? Transactions of the Association for Computational Linguistics **10**, 359–375 (2022)

[27] Fu, Y., Peng, H., Ou, L., Sabharwal, A., Khot, T.: Specializing smaller language models towards multi-step reasoning. arXiv preprint arXiv:2301.12726 (2023)

[28] Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., Zhou, D.: Self-consistency improves chain of thought reasoning in language models. arXiv preprint arXiv:2203.11171 (2022)

[29] Ho, N., Schmid, L., Yun, S.-Y.: Large language models are reasoning teachers. arXiv preprint arXiv:2212.10071 (2022)

[30] Magister, L.C., Mallinson, J., Adamek, J., Malmi, E., Severyn, A.: Teaching small language models to reason. arXiv preprint arXiv:2212.08410 (2022)

[31] Li, L.H., Hessel, J., Yu, Y., Ren, X., Chang, K.-W., Choi, Y.: Symbolic chain-of-thought distillation: Small models can also" think" step-by-step. arXiv preprint arXiv:2306.14050 (2023)

[32] Hsieh, C.-Y., Li, C.-L., Yeh, C.-K., Nakhost, H., Fujii, Y., Ratner, A., Krishna, R., Lee, C.-Y., Pfister, T.: Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. arXiv preprint arXiv:2305.02301 (2023)

[33] Wang, P., Chan, A., Ilievski, F., Chen, M., Ren, X.: PINTO: Faithful language reasoning using prompt-generated rationales. In: Workshop on Trustworthy and Socially Responsible Machine Learning, NeurIPS 2022 (2022). https://openreview.net/forum?id=7cYEP8WYME9

[34] Tian, Y., Han, Y., Chen, X., Wang, W., Chawla, N.: Tinyllm: Learning a small student from multiple large language models. ArXiv **abs/2402.04616** (2024)

[35] Zhang, Y., Pan, B., Ling, C., Hu, Y., Zhao, L.: Elad: Explanation-guided large language models active distillation. arXiv preprint arXiv:2402.13098 (2024)

[36] Howard, J., Ruder, S.: Universal language model fine-tuning for text classification. arXiv preprint arXiv:1801.06146 (2018)

[37] Hinton, G., Vinyals, O., Dean, J.: Distilling the knowledge in a neural network. stat **1050**, 9 (2015)

[38] Tang, R., Lu, Y., Liu, L., Mou, L., Vechtomova, O., Lin, J.: Distilling task-specific knowledge from bert into simple neural networks. arXiv preprint arXiv:1903.12136 (2019)

[39] Wang, S., Liu, Y., Xu, Y., Zhu, C., Zeng, M.: Want to reduce labeling cost? GPT-3 can help. In: Moens, M.-F., Huang, X., Specia, L., Yih, S.W.-t. (eds.) Findings of the Association for Computational Linguistics: EMNLP 2021, pp. 4195–4205. Association for Computational Linguistics, Punta Cana, Dominican Republic (2021). https://doi.org/10.18653/v1/2021.findings-emnlp.354 . https://aclanthology.org/2021.findings-emnlp.354

[40] Smith, R., Fries, J.A., Hancock, B., Bach, S.H.: Language models in the loop: Incorporating prompting into weak supervision. arXiv preprint arXiv:2205.02318 (2022)

[41] Arora, S., Narayan, A., Chen, M.F., Orr, L., Guha, N., Bhatia, K., Chami, I., Sala, F., Ré, C.: Ask me anything: A simple strategy for prompting language models. arXiv preprint arXiv:2210.02441 (2022)

[42] Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., Liu, P.J.: Exploring the limits of transfer learning with a unified text-to-text transformer. Journal of machine learning research **21**(140), 1–67 (2020)

[43] Talmor, A., Herzig, J., Lourie, N., Berant, J.: Commonsenseqa: A question answering challenge targeting commonsense knowledge. arXiv preprint arXiv:1811.00937 (2018)

[44] Armengol-Estapé, J., Carrino, C.P., Rodriguez-Penagos, C., Gibert Bonet, O., Armentano-Oller, C., Gonzalez-Agirre, A., Melero, M., Villegas, M.: Are multilingual models the best choice for moderately under-resourced languages? A comprehensive assessment for Catalan. In: Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021, pp. 4933–4946. Association for Computational Linguistics, Online (2021). https://doi.org/10.18653/v1/2021.findings-acl.437 . https://aclanthology.org/2021.findings-acl.437

[45] Chung, H.W., Hou, L., Longpre, S., Zoph, B., Tay, Y., Fedus, W., Li, E., Wang, X., Dehghani, M., Brahma, S., Webson, A., Gu, S.S., Dai, Z., Suzgun, M., Chen, X., Chowdhery, A., Narang, S., Mishra, G., Yu, A., Zhao, V., Huang, Y., Dai, A., Yu, H., Petrov, S., Chi, E.H., Dean, J., Devlin, J., Roberts, A., Zhou, D., Le, Q.V., Wei, J.: Scaling Instruction-Finetuned Language Models. arXiv (2022). https://doi.org/10.48550/ARXIV.2210.11416 . https://arxiv.org/abs/2210.11416

[46] Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., Liu, P.J.: Exploring the limits of transfer learning with a unified text-to-text transformer. Journal of Machine Learning Research **21**(140), 1–67 (2020)

[47] Hsieh, C.-Y., Li, C.-L., Yeh, C.-k., Nakhost, H., Fujii, Y., Ratner, A., Krishna, R., Lee, C.-Y., Pfister, T.: Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. In: Rogers, A., Boyd-Graber, J., Okazaki, N. (eds.) Findings of the Association for Computational Linguistics: ACL 2023, pp. 8003–8017. Association for Computational Linguistics, Toronto, Canada (2023). https://doi.org/10.18653/v1/2023.findings-acl.507 . https://aclanthology.org/2023.findings-acl.507

[48] Valmeekam, K., Olmo, A., Sreedharan, S., Kambhampati, S.: Large language models still can't plan (a benchmark for LLMs on planning and reasoning about change). In: NeurIPS 2022 Foundation Models for Decision Making Workshop (2022). https://openreview.net/forum?id=wUU-7XTL5XO

[49] Jain, S., Ma, X., Deoras, A., Xiang, B.: Self-consistency for open-ended generations. arXiv preprint arXiv:2307.06857 (2023)